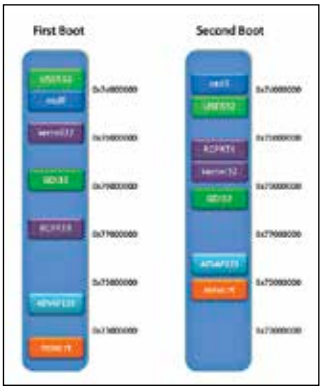

ASLR Randomizes the memory location of libraries each time you boot Windows 8

will either not load it in the first place (the strong configuration) or allow it to load at the memory location that the library prefers (weak configuration). A backport of this feature is available for Windows 7 at *http://support.microsoft.com/kb/2639308*. The update is also installed when you install Internet Explorer 10 on Windows 7.

HEASLR

Although ASLR tries to allay the possibilities of random code execution, its potential is limited on a 32-bit platform due to limited memory space available for allocation of libraries at the load-time. Theoretically 32-bit systems can use up to 4 GB of memory but Windows won't utilize more than 2 GB of it for any process' user address space. The ASLR algorithms in Windows Vista and Windows 7 until now have been using the same algorithm for both 32-bit and 64-bit platforms. This approach defeats the benefit ASLR could get on 64-bit systems with a lot more randomization space available. HEASLR or High Entropy ASLR utilizes the address space by giving more randomness to the memory addresses at which libraries are loaded in runtime thus elevating security to another level on 64-bit platforms. Like ASLR, HEASLR is also an optional feature.

If you're a developer, it's advisable to make sure that ASLR is enabled for the libraries that you create. While Visual Studio 2010 enables ASLR by default (make sure of it in `Project Properties > Configuration Properties > Linker > Advanced`), HEASLR needs to be enabled manually by adding the /HIGHENTROPYVA option to the linker.

Interestingly, Internet Explorer 10 is one of the first apps to take advantage of all the new ASLR improvements – and that makes it more secure than the competition against specific attacks.

Ivy Bridge and security enhancements to DEP

As we said earlier, DEP tries to enhance security by marking chunks of memory (called memory pages) as 'non-executable'. However, there are ways to bypass this. The first is by a return oriented programming attack,